# Theoretical Architecture, Axiomatic Foundations, and Formal Verification of the Bekenstein–Nyquist Shock Worldline Conjecture

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**Abstract**

The Bekenstein–Nyquist Shock Worldline Conjecture (BNSWC) establishes a rigorous mathematical bridge between bandlimited effective field theory (EFT), microlocal partial differential equation (PDE) analysis, and Boolean or unitary circuit complexity. Motivated by three foundational principles—the Bekenstein–Kempf spatial bandwidth cutoff ($\Lambda_{\mathrm{Bek}}$) [1, 2, 4], the Margolus–Levitin rate limit on local state transitions ($\nu_{\mathrm{ML}}$) [8, 7], and reflectionless local unitarity ($U^\dagger U = I$) [14, 15]—we formalize a bandlimited lattice EFT via four minimal axioms (Axioms A1–A4) under scale-generating nonlinearity ($[P_\Lambda, U_{\mathrm{nl}}] \neq 0$) [10]. Smooth, sub-Nyquist field configurations ($\Sigma(x,t) > 0$) propagate causally in $\mathcal{O}(1)$ parallel circuit depth per macro-step, whereas nonlinear convective steepening at a hydrodynamic shock front ($\Sigma(x,t) < 0$) saturates the local spatial bandwidth. Rather than postulating auxiliary degrees of freedom ad hoc, we prove via Stinespring and Sz.-Nagy dilation [17, 16] (`UnitaryBandwidthDepth.lean`) that preserving local unitarity without violating spatial bandlimitation mathematically forces the existence of an orthogonal worldline clock fiber $\mathcal{H}_{\mathrm{clock}} = V_\Lambda^\perp$ of rank at least $r = \mathrm{rank}([P_\Lambda, U_{\mathrm{nl}}])/2$, a 90° microlocal continuation in the complexified cotangent bundle across the sonic/Nyquist horizon $\Sigma(x,t) = 0$, and a localized Ginzburg–Landau shock-core defect-condensate potential $V_{\mathrm{eff}}(|c|^2) = \Sigma_0|c|^2 + \lambda|c|^4$ [30] that is algebraically isomorphic to the Standard Model Higgs potential [31, 34] under $\mu^2 = -\Sigma_0$. This monograph presents the unified axiomatic architecture, the five-phase analytical proofs, six machine-checked Lean 4 modules (`Mathlib`, `Physlib`, and `QuantumInfo`) [33, 34], four progressive hydrodynamic lattice benchmarks, the holographic Ryu–Takayanagi bulk-extrusion correspondence [28, 29], and the Ginzburg–Landau–Higgs defect-condensate duality.

## Contents

**Executive Synthesis: Motivating Physical Cutoffs and Deductive Architecture** **2**

**1 Axiomatic Foundations, Deductive Structure, and the Aliasing–Unitarity No-Go Lemma (Phase 1)** **3**
- 1.1 The Minimal Four-Axiom Formulation (Emergent Dilation Architecture) . . . . . 3
- 1.2 Deductive Emergence of the Clock Fiber, 90° Rotor, and Defect-Condensate Potential . . . . . 5
- 1.3 Axiomatic Equivalence: Emergent Dilation vs. Primitive Bipartite Bundle . . . . 5
- 1.4 Analytical Proof Architecture for the Nyquist–Lax Trichotomy . . . . . 6

**2 Microlocal Analysis of the 90° Causal Rotation (Phase 2)** **7**
- 2.1 The Bekenstein–Nyquist Saturation Functional and Mixed-Type PDEs . . . . . 7
- 2.2 Complex Coordinate Continuation and Internal Clock Winding . . . . . 9

**3 Complexity Theory and Sequential Depth Lower Bounds (Phase 3) 9**
3.1 The Sub-Stepping Depth Theorem and Holographic Shock Complexity . . . . . 9
3.2 Planar Circuit Value Problem (PCVP) Reduction, the Monotone Trap, and the $\pi$-Rotor Negation Gate . . . . . 9
3.3 Apparent Lax Shock Entropy vs. Zero Global Landauer Entropy . . . . . 10

**4 Computational Verification on Bandlimited Hydrodynamic Lattices (Phase 4) 11**
4.1 The $1+1$D Unitary Nyquist Rotor Lattice (Experiment H.3 / Milestone 2) . . . 11
4.2 Comparative Shock-Capturing Benchmarks and Poisson–Euler RG Duality . . . . 11
4.3 Spacetime Causal DAG Critical-Path Scaling and Asymptotic Tightness (Experiment H.4 / Milestone 3) . . . . . 12
4.4 The Bandwidth–Saturation Equivalence Principle and Holographic Bulk Extrusion 12
4.4.1 Holographic Bulk Extrusion, Ryu–Takayanagi Min-Cut Wire Counting, and Effective $g_{00}$ Horizon Geometry . . . . . 13
4.5 Two-Shock Collision Interferometry, Coherent Clock-Mass Annihilation, and the Ginzburg–Landau Defect Condensate (Milestone 4 & Phase 3.2) . . . . . 14
4.5.1 Evasion of the Monotone Planar Circuit Trap via Orthogonal Two-Mode Haar Clock Readback . . . . . 14
4.5.2 Experimental Verification: Gate Truth Table, Coalescence Kinematics, and Ramsey Interference Fringes . . . . . 14
4.5.3 Localized Ginzburg–Landau Shock-Core Defect Condensate and Isomorphism with the Higgs Potential . . . . . 14

**5 Machine-Checked Formal Verification in Lean 4 (Phase 5, Phase A & Phase B) 15**
5.1 Core Bandwidth–Depth Tradeoff (`UnitaryBandwidthDepth.lean`) . . . . . 15
5.2 Extended Physics Formalizations and Upstream `Physlib` / `QuantumInfo` Bridges 17

**6 Integrated Synthesis 19**

## Executive Synthesis: Motivating Physical Cutoffs and Deductive Architecture

The Bekenstein–Nyquist Shock Worldline Conjecture investigates a bandlimited lattice effective field theory (EFT) motivated by simultaneous enforcement of three foundational principles of fundamental physics that have historically been analyzed in isolation:

1. **The Bekenstein–Kempf Spatial Bandwidth Cutoff ($\Lambda_{\mathbf{Bek}}$):** Black hole thermodynamics and algebraic quantum field theory establish the Bekenstein bound [1, 2], restricting the maximum information $I_{\max}$ storable within a region of radius $R$ and energy $E$ to

$$I_{\max} \leq \frac{2\pi ER}{\hbar c \ln 2}. \tag{1}$$

Following Achim Kempf's covariant bandlimitation framework [3, 4, 5], we treat a finite local spatial information density as motivating an effective ultraviolet wavenumber cutoff $|k| \leq \Lambda_{\text{Bek}}$, restricting observable continuum fields to a Paley–Wiener Hilbert space $\mathcal{H}_\Lambda = PW_{\Lambda_{\text{Bek}}}(\mathbb{R})$ obeying Bernstein's inequality $\|u^{(r)}\|_2 \leq \Lambda_{\text{Bek}}^r \|u\|_2$. By the Shannon–Nyquist sampling theorem [6], $\mathcal{H}_\Lambda$ is unitarily isomorphic to a discrete sampling lattice $\ell^2(\mathbb{Z}_{\Delta x})$ of fundamental spacing $\Delta x = \pi/\Lambda_{\text{Bek}}$.

2. **The Margolus–Levitin Rate Limit on State Changes ($\nu_{\mathrm{ML}}$):** Quantum mechanics enforces the Mandelstam–Tamm [7] and Margolus–Levitin [8, 9] quantum speed limits, which bound the maximum frequency of distinguishable transitions between orthogonal quantum states by

$$\nu_{\mathrm{ML}} \leq \frac{2E}{\pi\hbar}, \qquad \Delta\tau_{\mathrm{ML}} = \frac{\pi}{\nu_{\mathrm{ML}}} \geq \frac{\pi^2\hbar}{2E}. \tag{2}$$

   Within our bandlimited lattice EFT, the spatial cutoff $\Lambda_{\mathrm{Bek}}$ and the local gate-rate ceiling $\nu_{\mathrm{ML}}$ act as fixed ultraviolet/rate parameters defining a local spacetime information-processing capacity per unit length per unit time:

$$C_{\mathrm{self}} \equiv \theta_{\mathrm{Nyq}} = \Lambda_{\mathrm{Bek}}\nu_{\mathrm{ML}}. \tag{3}$$

3. **Local Microscopic Unitarity vs. Landauer Erasure ($U^\dagger U = I$):** Fundamental quantum evolution is strictly unitary and causally local (bounded spatial fan-in radius $r_0$ per parallel step [19]), conserving total Hilbert-space norm and phase-space volume [15]. While linear dispersive waves propagate trivially inside $\mathcal{H}_\Lambda$, nonlinear convective steepening (such as $u\partial_x u$ in Burgers/Euler hydrodynamics [11, 13]) multiplies modes in the upper half of the Brillouin zone ($\Lambda_{\mathrm{Bek}}/2 < |k| \leq \Lambda_{\mathrm{Bek}}$), driving local ultraviolet spillover flux $\mathcal{F}_{tx}(x,t) = \|P_{\mathrm{UV}}(u\partial_x u)(x,t)\|$ toward the forbidden super-Nyquist zone $(\Lambda_{\mathrm{Bek}}, 2\Lambda_{\mathrm{Bek}}]$. Standard continuum hydrodynamics stabilizes shocks via viscous dissipation [11] (violating unitarity via Landauer bit erasure $E_{\min} = nk_BT\ln 2$ [14]), whereas naive unitary lattice truncation reflects ultraviolet modes back into the bulk via Umklapp aliasing (producing dispersive Gibbs/Lax–Levermore oscillations [12]).

The central thesis of this monograph is that a bandlimited unitary system resolves nonlinear shock steepening under these three constraints without Landauer erasure or Gibbs aliasing by dynamically trading **spatial parallelism** for **sequential temporal circuit depth** along the lower-dimensional shock worldline.

# 1 Axiomatic Foundations, Deductive Structure, and the Aliasing–Unitarity No-Go Lemma (Phase 1)

## 1.1 The Minimal Four-Axiom Formulation (Emergent Dilation Architecture)

To separate physical resource inputs from derived mathematical structures, we formulate the conjecture on a composite Hilbert space $\mathcal{H}_{\mathrm{tot}}$ governed by four primitive axioms: two kinematic resource cutoffs (Bekenstein–Kempf and Margolus–Levitin), one dynamical nonlinearity condition, and one local unitarity conservation law.

**Axiom 1.1** (Spatial Bandlimitation — The Bekenstein–Kempf Projector (A1))**.** Observable spatial fields reside in a Paley–Wiener subspace $V_\Lambda = P_\Lambda\mathcal{H}_{\mathrm{tot}} = PW_{\Lambda_{\mathrm{Bek}}}(\mathbb{R})$ [4, 5] defined by an idempotent, self-adjoint ultraviolet wavenumber projector $P_\Lambda^2 = P_\Lambda = P_\Lambda^\dagger$ with hard spectral cutoff $|k| \leq \Lambda_{\mathrm{Bek}}$, unitarily isomorphic to the discrete sampling lattice $\ell^2(\mathbb{Z}_{\Delta x})$ with fundamental spacing $\Delta x = \pi/\Lambda_{\mathrm{Bek}}$ [6].

**Axiom 1.2** (Rate Limit on State Changes — The Margolus–Levitin Sequential Channel (A2))**.** Any local state transition coupling the bandlimited spatial subspace $V_\Lambda$ to an orthogonal degree of freedom proceeds through elementary rank-1 quantum channels ($\mathrm{rank}((I - P_\Lambda)W_k P_\Lambda) \leq 1$) whose transition frequency is bounded above by the Margolus–Levitin rate parameter $\nu_{\mathrm{ML}} \leq \frac{2E}{\pi\hbar}$ [8, 7] (imposing a minimum sequential clock tick $\Delta\tau_{\mathrm{ML}} = \pi/\nu_{\mathrm{ML}}$ and maximum single-step rotation angle $\vartheta_{\max}$ per elementary gate $W_k$).

Table 1: Five-Phase Architecture of the Bekenstein–Nyquist Shock Worldline Program.

| **Program Phase** | **Core Objective** | **Mathematical Apparatus** | **Target Deliverable** | **Falsification Criterion** |
|---|---|---|---|---|
| **Phase 1: Axioms & Kinematics** | State Minimal Axioms A1–A4 and prove incompatibility of spatial parallelism and unitarity at a shock | Paley–Wiener spaces $\mathcal{H}_\Lambda$, discrete Lax pairs $(L, M)$, Hilbert–Schmidt commutator norms | Minimal Axiomatization & **Nyquist–Lax Trichotomy** | Local bounded-fan-in spatial stencil resolving a Lax spike without dissipation or aliasing |
| **Phase 2: Microlocal Dynamics** | Regularize loss of spatial hyperbolicity ($\Sigma < 0$) by $90^\circ$ continuation into clock rotors | Mixed-type Tricomi–Keldysh PDEs [21, 22], complexified cotangent bundle $T^*\mathcal{M}_\mathbb{C}$ | Continuation across $\Sigma(x,t) = 0$ from spatial hyperbolic ($\Sigma > 0$) into clock fiber $\tau$ ($\Sigma < 0$) | Failure of complex coordinate continuation to preserve symplectic/unitary norm |
| **Phase 3: Complexity Theory** | Translate shock sub-stepping into sequential circuit depth lower bounds | Bounded-depth circuits (NC vs. P), DAG critical-path analysis, communication complexity | **Sub-Stepping Depth Theorem** and $\pi$-rotor negation ($\neg$) gate for PCVP [23, 24] | Parallel prefix or spectral algorithms fast-forwarding intersecting shocks in $\mathcal{O}(\log^k N)$ depth |
| **Phase 4: Computational Sandboxes** | Benchmark unitary temporal sub-stepping against dissipative shock-capturing schemes | Bandlimited spectral lattices, unitary Haar–Givens rotors, co-moving clock registers in PyTorch | **Milestones 2, 3, & 4**: Zero-loss shock propagation, linear DAG scaling, coherent clock readback | Secular energy drift or non-physical dispersive reflection in temporal phase buffers |
| **Phase 5: Formal Verification** | Machine-check the finite-dimensional algebraic core and physics bridges in Lean 4 | Lean 4, `Mathlib`, `Physlib`, `QuantumInfo`, block-partitioned sum types [33, 34] | Six verified Lean 4 modules proving $D \geq \mathrm{rank}([P_\Lambda, U_{\mathrm{nl}}])/2$ and single-bond angle bounds | Failure of commutator rank or angle bounds to enforce a strict lower bound on sequential depth |

**Axiom 1.3** (Scale-Generating Nonlinearity — Non-Commutativity (A3))**.** The microscopic generator of spatial wave transport is nonlinear (convective mode-multiplying), so for steepening wave profiles the unconstrained short-time propagator $U_{\mathrm{nl}} = \exp(\Delta t M(u))$ fails to commute with the spatial bandlimit projector $P_\Lambda$ [10]:

$$[P_\Lambda, U_{\mathrm{nl}}] \neq 0 \quad \Longleftrightarrow \quad \mathcal{A}_\Lambda[u] \equiv \big\| \big[P_\Lambda, [M(u), L(u)]\big] \big\|_{\mathrm{HS}} > 0, \tag{4}$$

where $(L(u), M(u))$ denotes the Lax pair [10] of the nonlinear flow and $F_{tx}[u] = \partial_t M - \partial_x L + [M, L]$ is the zero-curvature defect, with pointwise ultraviolet spillover envelope density $\mathcal{F}_{tx}(x,t) = \|P_{\mathrm{UV}}(u\partial_x u)(x,t)\|$.

**Axiom 1.4** (Reflectionless Local Unitarity — No Erasure and No Aliasing (A4))**.** Time evolution on $\mathcal{H}_{\mathrm{tot}}$ is generated by bounded-fan-in local unitary operators ($U^\dagger U = I$, forbidding dissipative Landauer bit erasure $E_{\min} = n k_B T \ln 2$ [14, 15]) that satisfy reflectionless impedance matching at the Brillouin boundary $|k| = \Lambda_{\mathrm{Bek}}$ (forbidding unphysical Umklapp spectral folding $k \mapsto k - 2\Lambda_{\mathrm{Bek}}$ or backward dispersive Gibbs reflections into $V_\Lambda$).

## 1.2 Deductive Emergence of the Clock Fiber, 90° Rotor, and Defect-Condensate Potential

Under Axioms A1–A4, the auxiliary structures of the Bekenstein–Nyquist framework are not assumed ad hoc; they emerge as rigorous mathematical deductions along a single linear chain:

$$\underbrace{\text{Axioms A1 + A3}}_{\text{Bandlimit + Nonlinearity}} \implies \underbrace{[P_\Lambda, U_{\text{nl}}] \neq 0}_{\text{Phase 1 Defect}} \xrightarrow{+\text{A4}} \underbrace{V_\Lambda^\perp \equiv \mathcal{H}_{\text{clock}}}_{\text{Phase 5 Clock Fiber}} \xrightarrow{+\text{A2}} \underbrace{\mathcal{D}(\gamma)\ \&\ \Sigma(x,t) < 0}_{\text{Phases 2–4 Rotor \& Condensate}} . \tag{5}$$

Specifically:

1. **Deduction 1 — Existence of the Orthogonal Worldline Clock Fiber $\mathcal{H}_{\text{clock}}$ (From A1 + A3 + A4):** By Axiom A3 ($[P_\Lambda, U_{\text{nl}}] \neq 0$), the projected spatial update $\Phi_{\Delta t} = P_\Lambda U_{\text{nl}} P_\Lambda$ satisfies $(P_\Lambda U_{\text{nl}} P_\Lambda)^\dagger (P_\Lambda U_{\text{nl}} P_\Lambda) = P_\Lambda - A^\dagger A$ with inter-band leakage $A = (I - P_\Lambda) U_{\text{nl}} P_\Lambda \neq 0$, making $\Phi_{\Delta t}$ a strict contraction on $V_\Lambda$. By Axiom A4 (Unitarity) and the Sz.-Nagy/Stinespring dilation theorem [16, 17] (machine-checked in Phase 5, `UnitaryBandwidthDepth.lean`), restoring unitarity mathematically requires an orthogonal complement $V_\Lambda^\perp = (I - P_\Lambda)\mathcal{H}_{\text{tot}} \equiv \mathcal{H}_{\text{clock}}$ of dimension at least $r = \text{rank}(A) = \frac{1}{2}\text{rank}([P_\Lambda, U_{\text{nl}}])$.

2. **Deduction 2 — Sequential Depth Lower Bound $\mathcal{D}(\gamma)$ (From Deduction 1 + A2):** Combining the commutator rank identity $\text{rank}([P_\Lambda, U_{\text{nl}}]) = 2r$ with Axiom A2 forces any multi-wire unitary circuit realization $U_D = W_D \cdots W_1$ to have total gate count $D \geq r = \frac{1}{2}\text{rank}([P_\Lambda, U_{\text{nl}}])$ (`UnitaryBandwidthDepth.lean`), while on a single bond ($r = 1$) achieving a cumulative target Givens rotation angle $\Theta$ using elementary gates of step angle $\vartheta \leq \vartheta_{\max}$ requires $D \geq \lceil \Theta / \vartheta_{\max} \rceil$ sequential sub-steps (`single_bond_angle_substepping_bound` in `NyquistRotorGates.lean`), yielding the continuum Sub-Stepping Depth Theorem along the shock worldline $\gamma$ (Phase 3).

3. **Deduction 3 — Normalized Feasibility Margin $\Sigma(x,t)$ and the 90° Continuation (From A1–A4):** Identifying the local parallel channel capacity $C_{\text{self}} = \Lambda_{\text{Bek}} \nu_{\text{ML}}$ (from A1 and A2) and the pointwise nonlinear spectral spillover density $R_{\text{self}}(x,t) = \mathcal{F}_{tx}(x,t)$ (from A3) defines the dimensionless channel feasibility margin $\mathcal{M}(x,t)/C_{\text{self}}$, which is the Bekenstein–Nyquist Saturation Functional $\Sigma(x,t) = 1 - \frac{\mathcal{F}_{tx}(x,t)}{\Lambda_{\text{Bek}} \nu_{\text{ML}}}$ (Phase 2). When $\Sigma > 0$ ($R_{\text{self}} < C_{\text{self}}$), spatial acoustic propagation is causally hyperbolic on $\mathcal{H}_\Lambda$ and executes in $\mathcal{O}(1)$ parallel stencil depth per macro-step; when $\Sigma < 0$ ($R_{\text{self}} > C_{\text{self}}$), purely spatial hyperbolicity breaks down at gradient catastrophe and Axiom A4 forces overflowing modes into sequential phase winding in $\mathcal{H}_{\text{clock}}$ via the 90° continuation $x \mapsto x + i c_{\text{eff}} \tau$, $\Delta k \mapsto -i \Delta k$.

4. **Deduction 4 — Impedance-Matched Givens Rotor and Ginzburg–Landau Defect Condensate (From A3 + A4):** On the $2 \times 2$ subspace spanning the overflowing Nyquist bond mode $|u_{b,\text{UV}}\rangle \in V_\Lambda$ and the active clock wire $|c_{j,k}\rangle \in V_\Lambda^\perp$, reflectionless unitarity at $|k| = \Lambda_{\text{Bek}}$ (Axiom A4) fixes the SU(2) Givens rotation angle $\tan\vartheta = \sqrt{\Delta k / \Lambda_{\text{Bek}}}$. Because transferring norm $|c|^2$ into $\mathcal{H}_{\text{clock}}$ reduces the local spatial spillover overload linearly ($F(|c|) = F_0 - \kappa |c|^2$), the saturation functional relaxes as $\Sigma(|c|) = \Sigma_0 + 2\lambda |c|^2$, whose integral with respect to $|c|^2$ yields the localized Ginzburg–Landau shock-core defect-condensate potential $V_{\text{eff}}(|c|^2) = \Sigma_0 |c|^2 + \lambda |c|^4$ [30], algebraically isomorphic to the Standard Model Higgs potential [31, 32] under $\mu^2 = -\Sigma_0$ (Phase 4 & Phase B).

## 1.3 Axiomatic Equivalence: Emergent Dilation vs. Primitive Bipartite Bundle

Just as quantum mechanics admits mathematically equivalent formulations (the Schrödinger, Heisenberg, and path-integral pictures), the Bekenstein–Nyquist Conjecture can be axiomatized

either by deriving $\mathcal{H}_{\text{clock}}$ via Stinespring dilation [17] (Axioms A1–A4) or by taking the bipartite fiber bundle $\mathcal{H}_{\text{tot}} = \mathcal{H}_\Lambda \oplus \mathcal{H}_{\text{clock}}$ as a primitive geometric structure equipped with minimal horizon gating:

- **Axiom B1 (Global Bundle Unitarity):** Composite evolution $U_{\text{tot}}(\Delta t) \in \mathrm{U}(\mathcal{H}_\Lambda \oplus \mathcal{H}_{\text{clock}})$ conserves total norm $\|u\|_2^2 + \sum_{j,k} |c_{j,k}|^2 = \text{const}$.
- **Axiom B2 (Sub-Nyquist Dormancy / Horizon Gating):** The inter-bundle coupling $(I - P_\Lambda)U_{\text{tot}}P_\Lambda$ vanishes wherever $\mathcal{F}_{tx}(x,t) \le \Lambda_{\text{Bek}}\nu_{\text{ML}}$ ($\Sigma(x,t) \ge 0$).
- **Axiom B3 (Impedance-Matched Overflow Serialization):** Wherever $\Sigma(x,t) < 0$, overflowing ultraviolet modes in $\mathcal{H}_{2\Lambda} \ominus \mathcal{H}_\Lambda$ rotate into $\mathcal{H}_{\text{clock}}$ via sequential rank-1 SU(2) Givens gates at $\tan\vartheta = \sqrt{\Delta k/\Lambda_{\text{Bek}}}$, costing one Margolus–Levitin tick $\Delta\tau_{\text{ML}} = \pi/\nu_{\text{ML}}$ per gate.

**Proposition 1.1** (Theorem of Axiomatic Equivalence)**.** *The Emergent Stinespring Dilation Axioms (A1–A4) and the Primitive Bipartite Bundle Axioms (B1–B3) are operationally equivalent on all spatial observables in $V_\Lambda$ and induce identical sequential depth lower bounds $\mathcal{D}(\gamma)$, horizon metrics $\Sigma(x,t)$, and effective potentials $V_{\text{eff}}(|c|^2)$. Moreover, taking $\mathcal{H}_{clock}$ as a persistent co-moving bundle (B1–B3) naturally accommodates multi-step phase memory ($K = 64$ worldline clock slots in Milestone 4) and shock-core defect condensates ($|c_{eq}|^2 > 0$).*

Table 2: Three Complementary Perspectives on the Axiomatic Architecture of the Conjecture.

| **Axiomatic Lens** | **Core Mathematical Statement** | **Ontological Meaning of a Shock Wave** | **Ontological Meaning of Mass / Clock Depth** |
|---|---|---|---|
| **1. Operator-Algebraic / Homological Lens** (Mathematicians & Lean 4) | Non-commutativity of nonlinear evolution and bandlimitation: $[P_\Lambda, U_{\text{nl}}] \neq 0$ | Localized spike in the commutator obstruction $\mathcal{A}_\Lambda[u] = \|[P_\Lambda, [M, L]]\|_{\text{HS}}$ | Minimal Stinespring dilation rank $D \ge \text{rank}([P_\Lambda, U_{\text{nl}}])/2$ in $V_\Lambda^\perp$ |
| **2. Information-Theoretic / Queueing Lens** (Computer Scientists) | Zero-drop channel buffering with capacity $C_{\text{self}} = \Lambda_{\text{Bek}}\nu_{\text{ML}}$ and $U^\dagger U = I$ | Local channel overload ($\mathcal{M} < 0 \iff \Sigma < 0$) where flux $R_{\text{self}} > C_{\text{self}}$ | Sequential queueing lag $m_j\Delta\tau_{\text{ML}}$ in local worldline clock memory $\mathcal{C}_j$ |
| **3. Geometric / Equivalence Principle Lens** (Relativistic & Particle Physicists) | Covariant bandlimit invariance on complexified cotangent bundle $T^*\mathcal{M}_{\mathbb{C}}$ | Sonic/Nyquist horizon ($\Sigma(x,t) = 0$) where effective metric $g_{00} \leftrightarrow \Sigma$ flips sign | Ginzburg–Landau defect condensate $\|c_{\text{eq}}\|^2 = -\Sigma_0/(2\lambda)$ coiled in shock fiber |

## 1.4 Analytical Proof Architecture for the Nyquist–Lax Trichotomy

To prove that spatial parallelism breaks down at a shock front under Axioms A1, A3, and A4, notice that "local unitarity" purely within the spatial Hilbert space $\mathcal{H}_\Lambda$ combines two distinct operational requirements: norm conservation ($\|\Phi_{\Delta t}(u)\|_2 = \|u\|_2$) and bounded-fan-in spatial locality ($\mathcal{O}(1)$ parallel circuit depth within stencil radius $r_0$ [19]). Theorem 1.2 proves that at a super-critical shock front, these two halves of spatial local unitarity cannot coexist with the Bekenstein–Kempf bandlimit (Axiom A1), forcing the relaxation of $\mathcal{O}(1)$ parallel depth into sequential time-stepping governed by the Margolus–Levitin rate limit (Axiom A2).

**Theorem 1.2** (Nyquist–Lax Trichotomy / Aliasing–Unitarity No-Go Lemma)**.** *Any single-step spatial update operator $\Phi_{\Delta t} : \mathcal{H}_\Lambda \to \mathcal{H}_\Lambda$ acting across a shock locus where $\mathcal{A}_\Lambda[u] =$*

$\|[P_\Lambda, [M(u), L(u)]]\|_{HS} > \Theta_{DSW}$ *must violate at least one of the following three operational postulates:*

1. ***Postulate 1 — Strict Spatial Bandwidth (Bekenstein–Kempf Bound, Axiom A1):*** $\mathrm{supp}(\hat{u}) \subseteq [-\Lambda_{Bek}, \Lambda_{Bek}]$ *without dynamically allocating sub-Planckian spatial lattice nodes.*

2. ***Postulate 2 — Norm Unitarity / Information Conservation (First Half of Axiom A4):*** $\|\Phi_{\Delta t}(u)\|_2 = \|u\|_2$*, forbidding artificial viscosity, WENO truncation, or Landauer erasure [14] of ultraviolet modes.*

3. ***Postulate 3 — Constant Parallel Depth / Bounded-Fan-In Spatial Locality (Second Half of Local Unitarity):*** $\Phi_{\Delta t}$ *is computable in circuit depth* $\mathcal{O}(1)$ *purely within* $\mathcal{H}_\Lambda$ *with a spatial stencil fan-in bounded by a fixed sub-Nyquist radius* $r_0$*.*

*Proof Architecture.* Assume Postulates 1, 2, and 3 hold simultaneously on $\mathcal{H}_\Lambda$.

1. **Step 1 (Donoho–Stark Uncertainty Barrier):** By the Donoho–Stark uncertainty inequality [18] ($|T| \cdot |\Omega| \geq 1 - (\epsilon_T + \epsilon_\Omega)^2$), a shock discontinuity compressed to spatial width $\delta x \sim \pi/\Lambda_{\mathrm{Bek}}$ concentrates $\mathcal{O}(1)$ spectral weight at the Brillouin boundary $|k| \in [\Lambda_{\mathrm{Bek}}/2, \Lambda_{\mathrm{Bek}}]$, generating a non-zero inter-band leakage operator $A = (I - P_\Lambda)U_{\mathrm{nl}}P_\Lambda : \mathcal{H}_\Lambda \to \mathcal{H}_{2\Lambda} \ominus \mathcal{H}_\Lambda$.

2. **Step 2 (Unitary Defect Identity):** Decomposing the unconstrained unitary propagator $U_{\mathrm{nl}} = \exp(\Delta t M(u))$ on $\mathcal{H}_\Lambda \oplus \mathcal{H}_\Lambda^\perp$ yields $(P_\Lambda U_{\mathrm{nl}} P_\Lambda)^\dagger (P_\Lambda U_{\mathrm{nl}} P_\Lambda) = P_\Lambda - A^\dagger A$. Thus, any bandlimited update $\Phi_{\Delta t} = P_\Lambda U_{\mathrm{nl}} P_\Lambda$ dissipates norm $\|\Phi_{\Delta t} u\|_2^2 = \|u\|_2^2 - \|Au\|_2^2 < \|u\|_2^2$ (violating Postulate 2) whenever $Au \neq 0$.

3. **Step 3 (GNVW Chiral Index Obstruction and Group Velocity Reversal):** By the Gross–Nesme–Vogts–Werner (GNVW) index theorem for 1D Quantum Cellular Automata [20], any strictly local constant-depth unitary circuit restricted purely to $\mathcal{H}_\Lambda$ (Postulate 3) has a trivial inter-subspace flow index between $\mathcal{H}_\Lambda$ and $\mathcal{H}_{2\Lambda} \ominus \mathcal{H}_\Lambda$. Folding the chiral ultraviolet flux back into $[-\Lambda_{\mathrm{Bek}}, \Lambda_{\mathrm{Bek}}]$ via Umklapp aliasing ($k \mapsto k - 2\Lambda_{\mathrm{Bek}}$) reverses the group velocity $v_g = \partial\omega/\partial k < 0$ at $|k| = \Lambda_{\mathrm{Bek}}$ and produces non-physical backward dispersive ringing [12]. Preserving both Postulates 1 and 2 therefore requires Stinespring dilation [17] into an orthogonal register $\mathcal{H}_{\mathrm{clock}}$ across sequential depth $D \geq \mathrm{rank}(A)$ (violating Postulate 3) ticked at the Margolus–Levitin rate limit $\nu_{\mathrm{ML}}$ [8] (Axiom A2).

□

# 2 Microlocal Analysis of the 90° Causal Rotation (Phase 2)

## 2.1 The Bekenstein–Nyquist Saturation Functional and Mixed-Type PDEs

Once Phase 1 establishes that no constant-depth spatial operator on $\mathcal{H}_\Lambda$ can simultaneously preserve unitarity and bandwidth across a Lax curvature spike, Phase 2 constructs the dynamical escape mechanism forced by Axioms A1–A4: a microlocal continuation across the sonic/Nyquist horizon that rotates overflowing spatial modes by 90° into sequential temporal phase degrees of freedom governed by $\nu_{\mathrm{ML}}$.

**Definition 2.1** (Bekenstein–Nyquist Saturation Functional)**.** Let $\mathcal{F}_{tx}(x,t) = |w_{\mathrm{analytic}}(x,t)| = \|P_{\mathrm{UV}}(u\partial_x u)(x,t)\|$ denote the local spatial spillover envelope density of the zero-curvature defect $F_{tx}[u]$ (whose global spatial integral defines the Hilbert–Schmidt defect $\|F_{tx}\|_{\mathrm{HS}}$). The local

normalized feasibility margin $\mathcal{M}(x,t)/C_{\text{self}}$ of a bandlimited field $u(x,t) \in \mathcal{H}_\Lambda$ relative to the spacetime information-processing ceiling $\theta_{\text{Nyq}} = \Lambda_{\text{Bek}}\nu_{\text{ML}}$ is defined by

$$\Sigma(x,t) \equiv \frac{C_{\text{self}} - R_{\text{self}}(x,t)}{C_{\text{self}}} = 1 - \frac{\mathcal{F}_{tx}(x,t)}{\Lambda_{\text{Bek}}\,\nu_{\text{ML}}}. \tag{6}$$

On the spatial cotangent bundle $T^*\mathcal{M}$ with dual coordinates $(\xi,\omega)$ and characteristic bulk wave speed $c_0$, the effective bandlimited wave operator $P_{\text{eff}}$ has the generalized Tricomi–Keldysh [21, 22] principal symbol:

$$\sigma_{\text{eff}}(x,t;\xi,\omega) = \omega^2 - \Sigma(x,t)\,c_0^2\xi^2, \tag{7}$$

which governs three distinct microlocal regimes:

- **Sub-Nyquist Bulk ($\Sigma(x,t) > 0$ — Hadamard Well-Posed Spatial Hyperbolicity, $\mathcal{O}(1)$ Parallel Depth):** Where $\mathcal{F}_{tx}(x,t) < \Lambda_{\text{Bek}}\nu_{\text{ML}}$, $\sigma_{\text{eff}} = \omega^2 - \Sigma(x,t)c_0^2\xi^2$ has the physically causal **hyperbolic** acoustic signature $(-,+)$ with real spatial characteristic speeds $\frac{dx}{dt} = \pm\sqrt{\Sigma(x,t)}\,c_0$. The Cauchy initial-value problem is Hadamard well-posed and updates in $\mathcal{O}(1)$ parallel spatial stencil depth per macro-step ($m_j = 0$).

- **Sonic or Nyquist Horizon ($\Sigma(x,t) = 0$ — Parabolic Degeneracy Locus):** Along the codimension-1 caustic locus where $\mathcal{F}_{tx}(x,t) = \Lambda_{\text{Bek}}\nu_{\text{ML}}$, the effective spatial acoustic speed vanishes ($\sigma_{\text{eff}} = \omega^2$), forming a parabolic degeneracy surface analogous to a transonic sonic horizon [21, 22] or an effective horizon $g_{00} = 0$.

- **Super-Nyquist Shock Core ($\Sigma(x,t) < 0$ — Loss of Purely Spatial Hyperbolicity & 90° Clock Regularization):** Inside the shock core where $\mathcal{F}_{tx}(x,t) > \Lambda_{\text{Bek}}\nu_{\text{ML}}$, the $\xi^2$ coefficient flips sign so that purely on the spatial manifold $\mathcal{H}_\Lambda$, $\sigma_{\text{eff}} = \omega^2 + |\Sigma(x,t)|c_0^2\xi^2$ becomes **elliptic** with imaginary spatial characteristic speeds $\frac{dx}{dt} = \pm i\sqrt{|\Sigma(x,t)|}\,c_0$, signaling gradient catastrophe and Hadamard ill-posedness of unregularized bandlimited spatial transport. In the complexified cotangent fiber $T^*_{(x,t)}\mathcal{M}_{\mathbb{C}}$, applying the 90° Wick-type continuation $\Delta\xi \mapsto -i\Delta\xi$ paired with $x \mapsto x + ic_{\text{eff}}\tau$ restores causal hyperbolicity $\omega^2 - |\Sigma(x,t)|c_0^2\xi_\tau^2 = 0$ along the orthogonal internal clock fiber $\tau$!

Table 3: Mixed-Type PDE Universality Classes at the Nyquist Horizon $\Sigma(x,t) = 0$ [21, 22].

| **Mixed-Type Class** | **Canonical Normal Form near $x_\perp = 0$** | **Principal Symbol $\sigma(x_\perp;\xi_\perp,\omega)$** | **Characteristic Geometry at Horizon $x_\perp = 0$** | **Physical Role in Bekenstein–Nyquist Shocks** |
|---|---|---|---|---|
| **Tricomi Type** [21] | $\partial_t^2 u - x_\perp c_0^2 \partial_{x_\perp}^2 u = 0$ | $\omega^2 - x_\perp c_0^2 \xi_\perp^2$ | Characteristics $\frac{dx_\perp}{dt} = \pm c_0\sqrt{x_\perp}$ meet the horizon **orthogonally** in a semi-cubic cusp and reflect | Governs transverse **Rankine–Hugoniot phase reflections** across the shock zone boundaries |
| **Keldysh Type** [22] | $x_\perp \partial_t^2 u - c_0^2 \partial_{x_\perp}^2 u = 0$ | $x_\perp \omega^2 - c_0^2 \xi_\perp^2$ | Characteristics approach the horizon **tangentially** (asymptotically) without escaping into the bulk | Governs **worldline trapping** of overflowing UV modes along the shock trajectory |

### 2.2 Complex Coordinate Continuation and Internal Clock Winding

Across the turning point $\Sigma(x,t)=0$, applying complex coordinate continuation $x \mapsto x + ic_{\text{eff}}\tau$ paired with the 90° cotangent rotation $\Delta k \mapsto -i\Delta k$ maps an overflowing spatial mode $k = \Lambda_{\text{Bek}} + \Delta k$ ($\Delta k > 0$) into sequential temporal phase winding along the internal worldline clock fiber $\mathcal{C}_j$:

$$\exp(ikx) = \exp(i\Lambda_{\text{Bek}}x)\cdot\exp(i\Delta k\, x) \xrightarrow{x\mapsto x+ic_{\text{eff}}\tau,\ \Delta k\mapsto -i\Delta k} \exp(i\Lambda_{\text{Bek}}x)\cdot\exp(-i\omega_{\text{sub}}\tau), \quad (8)$$

where the internal sub-stepping frequency is $\omega_{\text{sub}} = c_{\text{eff}}\Delta k \le \nu_{\text{ML}}$.

## 3 Complexity Theory and Sequential Depth Lower Bounds (Phase 3)

### 3.1 The Sub-Stepping Depth Theorem and Holographic Shock Complexity

Phase 3 translates the sequential clock sub-stepping generated in the $\Sigma(x,t) < 0$ shock core into lower bounds on Boolean and unitary circuit depth, linking the PDE horizon transition directly to the separation between parallel complexity (NC) and sequential polynomial time (P).

**Theorem 3.1** (Sub-Stepping Depth Theorem). *For any bounded-precision unitary simulator evolving a field configuration along a shock worldline $\gamma$, the minimum sequential circuit depth $\mathcal{D}(\gamma)$ is bounded below by the line integral of the projection-curvature commutator anomaly:*

$$\mathcal{D}(\gamma) \ge \frac{1}{\pi\nu_{ML}}\int_{\gamma} \|[P_\Lambda,[M(u),L(u)]]\|_{HS}\, ds. \quad (9)$$

This theorem exhibits a direct structural parallel with the **Complexity = Action (CA)** and **Shock Wave Geometry** framework in holographic quantum gravity [25, 26, 27]. In holographic shock wave geometries [25], injecting shock waves into a bulk spacetime stretches the Wheeler–DeWitt (WDW) causal patch at a late-time rate governed by Lloyd's bound [9, 26], $\frac{d\mathcal{C}}{dt} \le \frac{2M}{\pi\hbar}$, while saturating the Maldacena–Shenker–Stanford (MSS) chaos bound $\lambda_L \le \frac{2\pi k_B T}{\hbar}$ [27]. Here, each unit of integrated commutator spillover flux requires at least one orthogonal Margolus–Levitin clock cycle to resolve unitarily. The proof strategy proceeds via **two-party communication complexity** across the moving shock cut $x_s(t)$ partitioning $\mathcal{H}_\Lambda$ into left ($\mathcal{H}_L$) and right ($\mathcal{H}_R$) Paley–Wiener subspaces: computing $K$ alternating non-commuting Rankine–Hugoniot unitary phase reflections $R_K R_{K-1}\cdots R_1$ across the cut requires $\Omega(K)$ sequential message-passing rounds, forbidding compression into polylogarithmic NC depth $\mathcal{O}(\log^k K)$.

### 3.2 Planar Circuit Value Problem (PCVP) Reduction, the Monotone Trap, and the $\pi$-Rotor Negation Gate

In $1+1$D spacetime, shock trajectories form a planar directed acyclic graph (DAG) of genus $g = 0$. While Goldschlager (1977) [23] proved that the general Planar Circuit Value Problem (PCVP) over $\{\wedge,\vee,\neg\}$ is P-complete, Limaye, Mahajan, and Sarma (2009) [24] proved that the **Monotone Planar Circuit Value Problem (MPCVP)** over $\{\wedge,\vee\}$ alone collapses into $\text{NC}^3$ ($\text{LogDCFL} \subseteq \text{NC}^2$ when stratified). Consequently, purely dissipative classical shocks whose amplitudes only add monotonically upon coalescence cannot escape $\text{NC}^3$. To break monotonicity in a planar $1+1$D shock network, colliding shock worldlines must realize a phase-coherent logical negation ($\neg$) gate via anti-phase ($\Delta\phi = \pi$) destructive clock interference $G_\pi(\vartheta)G_0(\vartheta) = I_2$ (demonstrated in Milestone 4 and machine-checked in `NyquistRotorGates.lean`). We note that while Milestone 4 explicitly realizes and verifies this unitary $\pi$-rotor negation ($\neg$) primitive, completing a full end-to-end P-completeness reduction for $1+1$D hydrodynamics remains a conjectured reduction requiring auxiliary fan-out (shock-splitting) and arrival-time synchronization gadgets.

Table 4: Circuit Value Problem (CVP) Variants and Implications for Shock Worldline Networks [23, 24].

| CVP Variant | Topology | Gate Set | Complexity Class | Shock Network Implication |
|---|---|---|---|---|
| **General CVP** | Arbitrary DAG (crossings allowed) | $\{\wedge, \vee, \neg\}$ | P-complete [23] | Requires crossover gadgets in $1+1$D multi-species or $2+1$D collisions |
| **Monotone CVP (MCVP)** | Arbitrary DAG (crossings allowed) | $\{\wedge, \vee\}$ only | P-complete [23] | Cannot be embedded in strictly planar $1+1$D single-species flow |
| **Planar CVP (PCVP)** | Planar DAG (genus $g = 0$) | $\{\wedge, \vee, \neg\}$ | P-complete [23] | **Target for Phase 3.2:** Milestone 4 demonstrates the unitary $\pi$-rotor $\neg$ gate; full reduction requires fan-out/sync gadgets |
| **Monotone Planar CVP (MPCVP)** | Planar DAG (genus $g = 0$) | $\{\wedge, \vee\}$ only | In $\mathrm{NC}^3$ [24] ($\mathrm{LogDCFL} \subseteq \mathrm{NC}^2$ stratified) | **The MPCVP Trap:** Dissipative monotone shock merging is parallelizable in $\mathcal{O}(\log^3 N)$ depth |

### 3.3 Apparent Lax Shock Entropy vs. Zero Global Landauer Entropy

A classical paradox in shock wave physics is the tension between Lax's macroscopic entropy inequality [11] (which requires irreversible kinetic energy loss $\Delta\mathcal{E}_{\text{spatial}} < 0$ across a compressive shock front to select the unique physically admissible weak solution) and microscopic quantum unitarity [14, 15] (which forbids irreversible bit erasure and guarantees zero global von Neumann entropy production). Within the Bekenstein–Nyquist bipartite architecture $\mathcal{H}_{\text{tot}}$, both principles hold simultaneously:

1. **Zero Global Landauer Entropy Production:** Because the joint fluid-plus-clock evolution $U_{\text{tot}}$ is strictly unitary, any pure initial state $|\Psi\rangle$ remains globally pure for all time, with global von Neumann entropy $S_{\text{vN}}(|\Psi\rangle\langle\Psi|) = 0$ (`bipartite_pure_global_entropy_zero` in `QuantumInfoClockBridge.lean`).

2. **Apparent Lax Entropy as Clock-Fiber Entanglement:** A coarse-grained observer who measures only the bandlimited spatial fluid $\rho_\Lambda = \mathrm{Tr}_C(|\Psi\rangle\langle\Psi|)$ traces out the orthogonal worldline clock fiber $\mathcal{H}_{\text{clock}}$, observing a strict spatial norm contraction $\|u(T)\|_2^2 = \|u(0)\|_2^2 - \sum_{j,k} |c_{j,k}(T)|^2$ and an apparent macroscopic shock entropy $S_{\text{vN}}(\rho_\Lambda) = S_{\text{vN}}(\rho_C) \geq 0$ (`spatial_shock_entropy_eq_clock_entropy`) with bipartite mutual information $I(\Lambda : C) = 2S_{\text{vN}}(\rho_\Lambda)$.

3. **Causal Bottleneck and Mori–Zwanzig Memory Kernel Equivalence:** In computational mechanics and statistical physics, physical laws simultaneously act as *information compression* (represented by the bandlimit/coarse-graining projector $P_\Lambda$, corresponding to Crutchfield's causal state bottleneck or $\epsilon$-machine [37, 38]) and as *predictive calculators* (represented by the compressed propagator $P_\Lambda U P_\Lambda$). Machine-checked in `UnitaryBandwidthDepth.lean` (`mori_zwanzig_markov_defect_eq_reverse_mul_forward` and `mori_zwanzig_markov_defect_rank_le`), the 2-step non-Markovian prediction defect satisfies the exact operator identity

$$P_\Lambda(WU)P_\Lambda - (P_\Lambda W P_\Lambda)(P_\Lambda U P_\Lambda) = \big(P_\Lambda W(I - P_\Lambda)\big)\big((I - P_\Lambda)UP_\Lambda\big) = B_W A_U, \quad (10)$$

whose rank is bounded above by both the forward Stinespring leakage rank $\mathrm{rank}(A_U)$ and the reverse leakage rank $\mathrm{rank}(B_W)$. Thus, autonomous Markovian prediction commutes with coarse-graining if and only if $[P_\Lambda, U] = 0$; when $[P_\Lambda, U] \neq 0$, the Mori–Zwanzig memory kernel [35, 36] required to close the macro-dynamics is literally the product $B_W A_U$ of the reverse and forward Stinespring clock-fiber leakage operators.

# 4 Computational Verification on Bandlimited Hydrodynamic Lattices (Phase 4)

## 4.1 The $1+1$D Unitary Nyquist Rotor Lattice (Experiment H.3 / Milestone 2)

To validate Phases 1 and 2 on a nonlinear hydrodynamic PDE without artificial viscosity or spectral aliasing, Experiment H.3 (`toy_model_2_nyquist_rotor_pde.py`) constructs the **Unitary Nyquist Rotor Lattice** in IEEE 754 `float64`/`complex128` precision on a periodic $1+1$D ring $x \in [-\pi, \pi)$ of $N_x = 2\Lambda_{\mathrm{Bek}}$ spatial nodes with fundamental Nyquist spacing $\Delta x = \pi/\Lambda_{\mathrm{Bek}}$. Each spatial site $j \in \{0, \dots, N_x - 1\}$ is augmented with an $M$-level local worldline clock register $\mathcal{C}_j = \mathrm{span}\{|c_{j,1}\rangle, \dots, |c_{j,M}\rangle\}$, yielding the composite Hilbert space $\mathcal{H}_{\mathrm{tot}} = \mathcal{H}_\Lambda \oplus \bigoplus_{j=0}^{N_x-1} \mathcal{C}_j$. The solver evolves the inviscid Burgers conservation law $\partial_t u + \partial_x(\frac{1}{2}u^2) = 0$ [11, 13] via three state-driven components:

1. **Translation-Invariant $\mathcal{H}_{2\Lambda}$ Analytic-Signal Spillover Sensor:** At each step, $u \in \mathcal{H}_\Lambda$ is zero-padded onto a $2\times$-oversampled Paley–Wiener grid $N_{\mathrm{fine}} = 4\Lambda_{\mathrm{Bek}}$ to evaluate $w_{\mathrm{fine}} = u_{\mathrm{fine}} \partial_x u_{\mathrm{fine}}$. Retaining the positive super-Nyquist band $\Lambda_{\mathrm{Bek}} \leq k_{\mathrm{fine}} < 2\Lambda_{\mathrm{Bek}}$ defines the complex analytic signal:

$$w_{\mathrm{analytic}}(x_{\mathrm{fine}}) = 2\,\mathcal{F}^{-1}\big(\mathbf{1}_{\{\Lambda_{\mathrm{Bek}} \leq k_{\mathrm{fine}} < 2\Lambda_{\mathrm{Bek}}\}}\,\mathcal{F}(u_{\mathrm{fine}}\partial_x u_{\mathrm{fine}})\big)\,. \tag{11}$$

   For a bandlimited step of jump $\Delta u$, $|w_{\mathrm{analytic}}|$ scales organically as $c_\infty \Lambda_{\mathrm{Bek}} (\Delta u)^2$ with $c_\infty \approx 0.264$ ($17.630, 34.580, 68.369$ at $\Lambda_{\mathrm{Bek}} = 64, 128, 256$), determining $m_j = \lceil F_j/\theta_{\mathrm{Nyq}} \rceil$.

2. **Skew-Symmetric Cayley Transport with Constitutive Subgrid Phase-Lag Closure:** Decomposing $u = \bar{u} + \delta u$ via a resolution-consistent binomial diffusion filter ($N_{\mathrm{pass}}(r) = N_0 r^2$ in the grid refinement ratio $r = (\pi/64)/\Delta x = \Lambda_{\mathrm{Bek}}/64$, so that the physical Gaussian smoothing width $\sigma_x = \sqrt{N_{\mathrm{pass}}/2}\,\Delta x$ remains invariant across $N_{\mathrm{pass}} = 4, 16, 64$ at $\Lambda_{\mathrm{Bek}} = 64, 128, 256$) applies a constitutive subgrid phase-lag closure $(1 + \gamma\tilde{\tau}_j)^{-1}$ to the wave-frame fluctuation $\delta u_j$:

$$w_{\mathrm{eff},j} = \frac{2}{3}\left(\bar{u}_j + \frac{u_j - \bar{u}_j}{1 + \gamma\,\tilde{\tau}_j}\right), \qquad U_{\mathrm{spatial}}(\Delta t) = \left(I + \frac{\Delta t}{2} A(w_{\mathrm{eff}})\right)^{-1} \left(I - \frac{\Delta t}{2} A(w_{\mathrm{eff}})\right), \tag{12}$$

   where $A(w)^T = -A(w)$, guaranteeing $U_{\mathrm{spatial}}^T U_{\mathrm{spatial}} = I$ to machine precision ($\sim 10^{-16}$).

3. **5-Bond Unitary Haar–Givens Rotor Sweep:** Each active shock node executes $m_j$ sequential $2 \times 2$ unitary Givens rotations $G_{j,k}(\vartheta_{j,k}, \phi_j) \in \mathrm{SU}(2)$ coupling the 2-site Haar Nyquist difference mode $u_{b,\mathrm{UV}} = (u_{b_0} - u_{b_1})/\sqrt{2}$ to a single 1D clock wire $c_{j,\mathrm{slot}} \in \mathcal{C}_j$ at the impedance-matched angle $\tan\vartheta = \sqrt{1/\Lambda_{\mathrm{Bek}}}$.

## 4.2 Comparative Shock-Capturing Benchmarks and Poisson–Euler RG Duality

Because the impedance-matched Givens coupling scales as $\sin^2\vartheta = \frac{1}{\Lambda_{\mathrm{Bek}}+1} \sim \Lambda_{\mathrm{Bek}}^{-1}$ while the spectral spillover sub-step count at a shock scales as $m_j = c(\Delta u)\Lambda_{\mathrm{Bek}}$, the per-macro-step

Table 5: Milestone 2 Comparative Benchmark Results ($\Lambda_{\text{Bek}} = 64, N_x = 128$).

| Scenario | Solver Architecture | Max Norm Drift | Spatial $\|u\|_2^2$ | Clock $\sum \|c\|^2$ | Final $TV(u)$ | Overshoot | Speed $s$ ($s_{\text{RH}}$) |
|---|---|---|---|---|---|---|---|
| Riemann Step | **Unitary Nyquist Rotor** | **1.245e-15** | 32.638543 | 1.611907 | 1.6616 | 2.612e-02 | 0.5727 (0.6000) |
| Riemann Step | Dissipative Godunov | 5.536e-02 | 32.354491 | 0.000000 | 1.5942 | 0.000e+00 | 0.6106 (0.6000) |
| Riemann Step | Naive Galerkin Spectral | 3.615e-07 | 34.250438 | 0.000000 | 10.1806 | 1.767e-01 | 0.6080 (0.6000) |
| Sine Wave $-\sin(x)$ | **Unitary Nyquist Rotor** | **6.661e-16** | 58.267303 | 5.732697 | 4.1198 | 2.638e-02 | 0.0004 (0.0000) |
| Sine Wave $-\sin(x)$ | Dissipative Godunov | 1.120e-01 | 56.831485 | 0.000000 | 3.8780 | 0.000e+00 | 0.0000 (0.0000) |
| Sine Wave $-\sin(x)$ | Naive Galerkin Spectral | 4.590e-08 | 63.999997 | 0.000000 | 33.9979 | 3.446e-01 | 0.0000 (0.0000) |

ultraviolet energy absorption converges to a cutoff-independent Poisson–Euler renormalization-group fixed point:

$$\mathcal{T}_{\text{UV}}(\Lambda_{\text{Bek}}) = 1 - (\cos^2\vartheta)^{m_j} = 1 - \left(1 - \frac{1}{\Lambda_{\text{Bek}} + 1}\right)^{c(\Delta u)\,\Lambda_{\text{Bek}}} \xrightarrow{\Lambda_{\text{Bek}} \to \infty} 1 - e^{-c(\Delta u)}. \tag{13}$$

### 4.3 Spacetime Causal DAG Critical-Path Scaling and Asymptotic Tightness (Experiment H.4 / Milestone 3)

Experiment H.4 (`toy_model_3_dag_complexity.py`) dynamically constructs the spacetime causal dependency DAG $G = (V, E)$ across bandwidth cutoffs $\Lambda_{\text{Bek}} \in \{64, 128, 256\}$ ($T_{\text{final}} = 1.60, \Delta t = 0.01$) for (a) a single-shock sine wave $u_0(x) = -\sin(x)$ ($t^* = 1.00$) and (b) a 4-mode random multi-shock gas ($t^* = 0.80$). At each macro-step $\Delta t$, $m_j$ is partitioned across $n_{\text{sub}}$ Cayley sub-steps via $m_{\text{sub},s} = \lfloor (m_j + s)/n_{\text{sub}} \rfloor$ so that the cumulative physical Givens rotations match the DAG rotor vertices 1-to-1:

$$\text{total_physical_givens_rotations} = \sum_{n=1}^{N_{\text{steps}}} \sum_{j=0}^{N_x - 1} m_j(t_n) = |V(T)| - (N_{\text{steps}} + 1)N_x. \tag{14}$$

Along the critical shock worldline $j_{\text{crit}}(t_n)$, the simulation tracks both (a) the **Unrounded Discrete Gate-Sum Lower Bound** $\mathcal{B}_{\text{thm}} = \sum_{n=1}^{N_{\text{steps}}} F_{j_{\text{crit}}}(t_n)/\theta_{\text{Nyq}}$ (verifying that integer ceiling sub-stepping $m_j = \lceil F_j/\theta_{\text{Nyq}} \rceil$ converges to the unrounded flux demand as $1 + \mathcal{O}(\Lambda_{\text{Bek}}^{-1})$) and (b) the **Physical Time-Integral Flux** $\mathcal{I}_{\text{flux}}(T) = \int_0^T F_{j_{\text{crit}}}(t)\,dt = \sum_{n=1}^{N_{\text{steps}}} F_{j_{\text{crit}}}(t_n)\Delta t = (\Delta t\,\theta_{\text{Nyq}})\,\mathcal{B}_{\text{thm}}$.

Table 6: Milestone 3 Spacetime Causal DAG Critical-Path Scaling and Theorem Tightness.

| Initial Condition | $\Lambda_{\text{Bek}}$ | Pre-Shock | Final depth($G$) | Excess $D_{\text{excess}}$ | Gate-Sum $\mathcal{B}_{\text{thm}}$ | Tightness | Work $\|V\|$ | Rotations $\sum m_j$ | Hyp. Slope | Max Drift |
|---|---|---|---|---|---|---|---|---|---|---|
| Single-Shock $-\sin(x)$ | 64 | 50 | 1014 | 854 | 816.19 | **1.0463** | 27086 | 6478 | 1637.04 | 6.661e-16 |
| Single-Shock $-\sin(x)$ | 128 | 50 | 2591 | 2431 | 2403.53 | **1.0114** | 62147 | 20931 | 4538.89 | 1.332e-15 |
| Single-Shock $-\sin(x)$ | 256 | 50 | 5605 | 5445 | 5417.13 | **1.0051** | 146927 | 64495 | 10088.89 | 1.998e-15 |
| Multi-Shock Gas | 64 | 40 | 541 | 381 | 329.83 | **1.1551** | 25545 | 4937 | 609.46 | 4.152e-16 |
| Multi-Shock Gas | 128 | 40 | 1193 | 1033 | 995.73 | **1.0374** | 56740 | 15524 | 1483.78 | 9.688e-16 |
| Multi-Shock Gas | 256 | 40 | 2293 | 2133 | 2118.86 | **1.0067** | 129990 | 47558 | 2963.51 | 9.688e-16 |

### 4.4 The Bandwidth–Saturation Equivalence Principle and Holographic Bulk Extrusion

The analytical and numerical structure of Phases 2–5 establishes an operational **Bandwidth–Saturation Equivalence Principle**:

1. **Informational Horizon Geometry and Proper-Time Dilation ($\Sigma(x,t)$ as an Effective Metric Component $g_{00}$):** $\Sigma(x,t) = 1 - \frac{\mathcal{F}_{tx}(x,t)}{\Lambda_{\text{Bek}}\nu_{\text{ML}}}$ enters $\sigma_{\text{eff}}(x,t;\xi,\omega) = \omega^2 - \Sigma(x,t)c_0^2\xi^2$ in structural analogy to the Schwarzschild temporal metric component $g_{00}(r) = 1 - \frac{2GM}{c^2 r}$, inducing a proper-time queueing lag $\Delta\tau_{\text{lag}}(\gamma) = \mathcal{D}(\gamma)\Delta\tau_{\text{ML}}$ across $\Sigma < 0$.

2. **Inertial Mass as Sequential Queueing Drag:** In the Galilean-invariant effective transport velocity $w_{\mathrm{eff},j} = \frac{2}{3}\big(\bar{u}_j + \frac{\delta u_j}{1+\gamma_{\mathrm{eff}}(r)\tilde{\tau}_j}\big)$ with $\tau_j = m_j \Delta\tau_{\mathrm{ML}}/\Delta t_{\mathrm{ref}}$, the constitutive closure factor $M_{\mathrm{inert},j} \equiv 1+\gamma_{\mathrm{eff}}(r)\tilde{\tau}_j$ acts as a state-dependent inertial drag multiplier decelerating the over-steepened wave crest to the Rankine–Hugoniot shock speed $s = [f(u)]/[u]$ [11].

3. **Rest Mass as Coiled Ultraviolet Momentum in the Clock Fiber:** In the stationary sine-wave shock benchmark ($u_0(x) = -\sin(x)$ at $T = 1.35 > t^* = 1.00$), the shock worldline sits macroscopically at rest at $x = 0$ ($s_{\mathrm{RH}} = 0.0000$, measured speed 0.0004) while storing $\mathcal{M}_{\mathrm{clock}} = \sum_{j,k} |c_{j,k}|^2 = 5.732697$ of ultraviolet $L^2$ norm in the stationary clock register $\mathcal{C}_{j_{\mathrm{shock}}}$ ticking at $\omega_{\mathrm{sub}} = c_{\mathrm{eff}}\Delta k \le \nu_{\mathrm{ML}}$.

#### 4.4.1 Holographic Bulk Extrusion, Ryu–Takayanagi Min-Cut Wire Counting, and Effective $g_{00}$ Horizon Geometry

Combining the Bekenstein–Nyquist Saturation Functional $\Sigma(x,t)$ of Phase 2 with the machine-checked Stinespring dilation bound of Phase 5 (`UnitaryBandwidthDepth.lean`) reveals a dynamical mechanism in which both an emergent orthogonal bulk/fiber dimension ($\mathcal{H}_{\mathrm{clock}}$) and an effective horizon metric ($g_{00} \leftrightarrow \Sigma(x,t)$) are extruded by ultraviolet spectral overflow on the spatial boundary $\mathcal{H}_\Lambda$ [28, 29].

**Corollary 4.1** (Boundary Overflow Drives Bulk Depth and Horizon Geometry)**.** *Let $u(x,t) \in \mathcal{H}_\Lambda = PW_{\Lambda_{Bek}}(\mathbb{R})$ evolve on a $1+1D$ bandlimited spatial lattice with Nyquist cutoff $\Lambda_{Bek}$, Margolus–Levitin transition ceiling $\nu_{ML}$, and composite Hilbert space $\mathcal{H}_{tot} = \mathcal{H}_\Lambda \oplus \bigoplus_{j=0}^{N_x-1} \mathcal{C}_j$. Then:*

1. ***Dynamical Extrusion of the Orthogonal Fiber:*** *In the sub-Nyquist causally hyperbolic bulk ($\Sigma(x,t) > 0$, $F_j \le \theta_{Nyq} = \Lambda_{Bek}\nu_{ML}$), the spectral spillover into $\mathcal{H}_{2\Lambda} \ominus \mathcal{H}_\Lambda$ vanishes exponentially, leaving the internal clock fiber unexcited ($m_j = 0$) and the circuit depth flat ($\mathcal{O}(1)$). When nonlinear convective steepening breaches the Nyquist horizon ($\Sigma(x,t) < 0$, $F_j > \theta_{Nyq}$), the complex coordinate continuation $x \mapsto x + ic_{eff}\tau$ paired with the $90^\circ$ cotangent rotation $\Delta k \mapsto -i\Delta k$ rotates overflowing boundary modes $\Delta k = k - \Lambda_{Bek} > 0$ into sequential temporal phase winding $\exp(-i\omega_{sub}\tau)$ along $\mathcal{C}_j$, dynamically extruding an internal fiber of sequential depth $m_j = \lceil F_j/\theta_{Nyq} \rceil \propto \Lambda_{Bek}(\Delta u)^2$ localized along the $1D$ shock worldline $\gamma$.*

2. ***Operator-Algebraic Ryu–Takayanagi / HRT Min-Cut Bound:*** *Each local Haar–Givens update decomposes a lattice bond $(u_j, u_{j+1})$ into its coarse-grained mean $\bar{u}_j = (u_j + u_{j+1})/\sqrt{2}$ and Nyquist difference mode $u_{j,UV} = (u_j - u_{j+1})/\sqrt{2}$, acting as a local multiscale wavelet disentangler (analogous to a MERA tensor network [29]) rotated into sequential Margolus–Levitin proper time. By Theorem `unitary_bandwidth_depth_tradeoff_clock_wires`, any unitary factorization $U_D = W_D \cdots W_1$ whose elementary gates $W_k$ couple $\mathcal{H}_\Lambda$ to at most a single $1D$ clock wire $C_k \subset \mathcal{H}_\Lambda^\perp$ ($\dim C_k = 1$) must sever a minimum number of clock wires along the worldline cut [28] bounded by half the commutator rank:*

$$\begin{aligned} \operatorname{rank}\big([P_\Lambda, U_{nl}]\big) = 2r \implies D \ge r = \operatorname{rank}\big((I - P_\Lambda)U_{nl}P_\Lambda\big), \\ \mathcal{D}(\gamma) \ge \frac{1}{\pi\nu_{ML}} \int_\gamma \|[P_\Lambda, [M(u), L(u)]]\|_{HS}\, ds. \end{aligned} \tag{15}$$

3. ***Apparent Shock Entropy and Effective $g_{00}$ Metric Warping:*** *Because total evolution on $\mathcal{H}_{tot}$ is strictly unitary ($\|u\|_2^2 + \sum_{j,k}|c_{j,k}|^2 = \mathrm{const}$), macroscopic hydrodynamic entropy production is the entanglement/norm leakage $\mathcal{M}_{clock} = \sum_{j,k}|c_{j,k}|^2$ obtained by tracing out the worldline clock fiber $\mathcal{H}_{clock}$ across $\gamma$. Under impedance matching $\sin^2\vartheta = (\Lambda_{Bek}+1)^{-1}$, the per-step leakage along $\gamma$ converges to the scale-invariant fixed point $1 - (\cos^2\vartheta)^{m_j} \to$*

$1-e^{-c(\Delta u)}$, *while the sequential queueing delay* $\Delta\tau_j = m_j\Delta\tau_{ML}$ *warps the effective principal symbol* $\sigma_{\mathit{eff}}(x,t;\xi,\omega) = \omega^2 - \Sigma(x,t)c_0^2\xi^2$ *in structural correspondence with the temporal metric component* $g_{00} \leftrightarrow \Sigma(x,t) = 1 - \frac{\mathcal{F}_{tx}}{\Lambda_{Bek}\nu_{ML}}$.

## 4.5 Two-Shock Collision Interferometry, Coherent Clock-Mass Annihilation, and the Ginzburg–Landau Defect Condensate (Milestone 4 & Phase 3.2)

### 4.5.1 Evasion of the Monotone Planar Circuit Trap via Orthogonal Two-Mode Haar Clock Readback

To demonstrate that intersecting $1+1$D shock worldlines physically realize the non-monotone logical negation ($\neg$) gate primitive required to break dissipative monotonicity [24], Milestone 4 (`toy_model_4_shock_collision_gate.py`) replaces the write-only pointer advance with a **co-moving $K$-slot resonant worldline clock register** ($K = 64$, indexed directly by slot $=$ (clock_ptr $+ k_{\mathrm{sub}}$) mod 64). On each active bond $(b_0, b_1)$, `execute_bidirectional_givens_collision` transforms $(u_{b_0}, u_{b_1})$ into the orthonormal 2-site Haar wavelet basis:

$$\bar{u}_{\mathrm{in}} = \frac{u_{b_0} + u_{b_1}}{\sqrt{2}} \quad \text{(smooth local mean)}, \qquad u_{\mathrm{UV,\,in}} = \frac{u_{b_0} - u_{b_1}}{\sqrt{2}} \quad \text{(Nyquist difference)}, \tag{16}$$

and executes a three-stage orthogonal update on the 3-mode subspace $(\bar{u}_b, u_{b,\mathrm{UV}}, c_{\mathrm{slot}})$:

1. **Stage 1 — Unitary UV Stabilization ($u_{b,\mathbf{UV}} \leftrightarrow c_{\mathbf{slot}}$):** Rotates the overflowing Nyquist difference mode $u_{\mathrm{UV,\,in}} > 0$ into the worldline clock slot $c_{\mathrm{slot}}$ by $\vartheta_1 = \min\big(\vartheta,\, 0.85\arctan(u_{\mathrm{UV,\,in}}/\max(0, c_{\mathrm{in}}))\big)$:

$$\begin{pmatrix} u_{\mathrm{UV,\,out}} \\ c_{\mathrm{mid}} \end{pmatrix} = \begin{pmatrix} \cos\vartheta_1 & -\sin\vartheta_1 \\ \sin\vartheta_1 & \cos\vartheta_1 \end{pmatrix} \begin{pmatrix} u_{\mathrm{UV,\,in}} \\ c_{\mathrm{in}} \end{pmatrix}. \tag{17}$$

2. **Stage 2 — Phase-Coherent Collision Readback into the Smooth Mean Mode ($\bar{u}_b \leftrightarrow c_{\mathbf{slot}}$):** When the state-driven shock separation crosses $\Delta x_{\mathrm{sep}} = x_2 - x_1 \le 2.4\Delta x$, the relative rotor phase $\Delta\phi = \phi_2 - \phi_1$ governs $\gamma_{\mathrm{phase}}(\Delta\phi) = \mathrm{sgn}(\sin(\Delta\phi/2))\sin^2(\Delta\phi/2)$ and $\theta_{\mathrm{rec}} = \min\big(\theta_{\mathrm{cap}},\, 0.40\gamma_{\mathrm{phase}}\arctan(c_{\mathrm{mid}}/\bar{u}_{\mathrm{in}})\big)$ (gated by a constitutive monotonicity-preserving TVD/MP plateau limiter derived from the smooth initial envelope bounds), rotating stored clock condensate $c_{\mathrm{mid}}$ back into the smooth local mean mode $\bar{u}_{\mathrm{out}}$ without exciting $u_{\mathrm{UV,\,out}}$:

$$\begin{pmatrix} \bar{u}_{\mathrm{out}} \\ c_{\mathrm{out}} \end{pmatrix} = \begin{pmatrix} \cos\theta_{\mathrm{rec}} & \sin\theta_{\mathrm{rec}} \\ -\sin\theta_{\mathrm{rec}} & \cos\theta_{\mathrm{rec}} \end{pmatrix} \begin{pmatrix} \bar{u}_{\mathrm{in}} \\ c_{\mathrm{mid}} \end{pmatrix}. \tag{18}$$

3. **Stage 3 — Inverse Orthogonal Haar Transformation:** Maps $(\bar{u}_{\mathrm{out}}, u_{\mathrm{UV,\,out}})$ back to $(u_{b_0}, u_{b_1})$:

$$\begin{aligned} u_{b_0}^{\mathrm{out}} &= u_{b_0}^{\mathrm{in}} + \frac{\bar{u}_{\mathrm{out}} - \bar{u}_{\mathrm{in}}}{\sqrt{2}} + \frac{u_{\mathrm{UV,\,out}} - u_{\mathrm{UV,\,in}}}{\sqrt{2}}, \\ u_{b_1}^{\mathrm{out}} &= u_{b_1}^{\mathrm{in}} + \frac{\bar{u}_{\mathrm{out}} - \bar{u}_{\mathrm{in}}}{\sqrt{2}} - \frac{u_{\mathrm{UV,\,out}} - u_{\mathrm{UV,\,in}}}{\sqrt{2}}. \end{aligned} \tag{19}$$

### 4.5.2 Experimental Verification: Gate Truth Table, Coalescence Kinematics, and Ramsey Interference Fringes

### 4.5.3 Localized Ginzburg–Landau Shock-Core Defect Condensate and Isomorphism with the Higgs Potential

Because Stage 1 transfers ultraviolet energy $|c|^2$ from $u_{b,\mathrm{UV}}$ into the clock fiber, the active commutator defect decreases linearly as $F(|c|) = F_0 - \kappa|c|^2$, shifting the instantaneous saturation

Table 7: Milestone 4 Suite B Two-Shock Collision & Coalescence Benchmark ($T_{\text{final}} = 1.40, \Lambda_{\text{Bek}} = 64$).

| Collision Solver / Regime | $\Delta\phi$ | Clock Mass $\mathcal{M}_{\text{clock}}(T)$ | Spatial $\|u(T)\|_2^2$ | Max $\|u\|$ | Final $TV(u)$ | Max Drift | Speed $s_{\text{merged}}$ | Rotations $\sum m_j$ |
|---|---|---|---|---|---|---|---|---|
| Unitary Rotor — In-Phase | 0 | 3.017201 | 49.695025 | 1.2433 | 2.8108 | **8.088e-16** | 0.6810 (-2.71%) | 1449 == 1449 |
| Unitary Rotor — Anti-Phase | $\pi$ | **0.693300** | **52.018926** | **1.3097** | **3.2517** | **8.088e-16** | 0.7678 (+9.69%) | 1565 == 1565 |
| Dissipative Godunov / WENO | — | 0.000000 | 49.155968 | 1.2000 | 2.0000 | 6.746e-02 | 0.7000 | 0 |

Table 8: Milestone 4 Suite C Shock Mach–Zehnder / Ramsey Interference Fringe ($\Delta\phi \in [0, 2\pi]$).

| $\Delta\phi$ | $\Delta\phi$ (deg) | $\mathcal{M}_{\text{clock}}(T, \Delta\phi)$ | $\|u(T, \Delta\phi)\|_2^2$ | $\|u\|_2^2 + \mathcal{M}_{\text{clock}}$ | Max $\|u(x,T)\|_\infty$ | Final $TV(u)$ | $\sum m_j$ | Max Drift |
|---|---|---|---|---|---|---|---|---|
| 0 | 0° | 3.017201 | 49.695025 | 52.712226 | 1.2433 | 2.8108 | 1449 | 8.088e-16 |
| $\pi/4$ | 45° | 2.594283 | 50.117943 | 52.712226 | 1.2505 | 3.3489 | 1450 | 8.088e-16 |
| $\pi/2$ | 90° | 1.141278 | 51.570949 | 52.712226 | 1.3049 | 2.8684 | 1581 | 8.088e-16 |
| $3\pi/4$ | 135° | 0.731896 | 51.980331 | 52.712226 | 1.3738 | 3.7672 | 1424 | 8.088e-16 |
| $\pi$ | 180° | **0.693300** | **52.018926** | 52.712226 | **1.3097** | **3.2517** | **1565** | 8.088e-16 |
| $5\pi/4$ | 225° | 0.731896 | 51.980331 | 52.712226 | 1.3738 | 3.7672 | 1424 | 8.088e-16 |
| $3\pi/2$ | 270° | 1.141278 | 51.570949 | 52.712226 | 1.3049 | 2.8684 | 1581 | 8.088e-16 |
| $7\pi/4$ | 315° | 2.594283 | 50.117943 | 52.712226 | 1.2505 | 3.3489 | 1450 | 8.088e-16 |
| $2\pi$ | 360° | 3.017201 | 49.695025 | 52.712226 | 1.2433 | 2.8108 | 1449 | 8.088e-16 |

functional to

$$\Sigma(|c|) = 1 - \frac{F_0 - \kappa|c|^2}{\Lambda_{\text{Bek}}\,\nu_{\text{ML}}} = \Sigma_0 + 2\lambda|c|^2, \qquad \lambda \equiv \frac{\kappa}{2\Lambda_{\text{Bek}}\nu_{\text{ML}}} > 0. \tag{20}$$

Integrating $\frac{\partial V_{\text{eff}}}{\partial(|c|^2)} = \Sigma(|c|)$ yields the effective Ginzburg–Landau [30] quartic free-energy functional governing the localized shock-core defect condensate:

$$V_{\text{eff}}(|c|^2) = \Sigma_0|c|^2 + \lambda|c|^4 = \underbrace{\left(1 - \frac{F_0(x,t)}{\Lambda_{\text{Bek}}\,\nu_{\text{ML}}}\right)}_{-\mu^2(x,t) \equiv \Sigma_0(x,t)} |c|^2 + \lambda|c|^4, \tag{21}$$

where in the smooth sub-Nyquist bulk ($\Sigma_0 > 0$) the clock fiber remains uncondensed ($|c_{\text{eq}}|^2 = 0$), while inside over-steepened shock cores ($\Sigma_0 < 0$) a localized defect condensate $|c_{\text{eq}}|^2 = -\Sigma_0/(2\lambda) > 0$ forms spontaneously to restore the sonic/Nyquist horizon $\Sigma(|c_{\text{eq}}|^2) = 0$. Under the sign identification $\mu^2 = -\Sigma_0$, $V_{\text{eff}}(|c|^2)$ is algebraically isomorphic to the Standard Model Higgs potential [31, 32, 34], with restoring curvature $\frac{1}{2}\frac{\partial^2 V_{\text{eff}}}{\partial|c|^2}\big|_{|c|=|c_{\text{eq}}|} = -2\Sigma_0$ measuring the stiffness of the Nyquist horizon $\Sigma = 0$.

# 5 Machine-Checked Formal Verification in Lean 4 (Phase 5, Phase A & Phase B)

## 5.1 Core Bandwidth–Depth Tradeoff (`UnitaryBandwidthDepth.lean`)

In `UnitaryBandwidthDepth.lean` (verified against Lean 4 v4.33.1 and `Mathlib` [33] with zero `sorry` axioms and zero compiler warnings), we formalize the finite-dimensional Hilbert space $H \cong \mathbb{C}^{d+m}$ indexed by the canonical block sum type `BlockIdx d m := Fin d ⊕ Fin m`. The orthogonal bandlimit projector onto the $d$-dimensional low-frequency Bekenstein subspace $V_\Lambda$ and its $m$-dimensional ultraviolet complement $V_\Lambda^\perp$ ($m = N - d$) are defined via `Matrix.fromBlocks` as $P_\Lambda = \text{diag}(I_d, 0_m)$ (`bandlimitProj d m`) and $I - P_\Lambda = \text{diag}(0_d, I_m)$ (`uvProj d m`):

$$\begin{aligned} P_\Lambda &= \begin{pmatrix} I_d & 0 \\ 0 & 0 \end{pmatrix}, \qquad U_{\text{nl}} = \begin{pmatrix} U_{11} & U_{12} \\ U_{21} & U_{22} \end{pmatrix}, \qquad A = \begin{pmatrix} 0 & 0 \\ U_{21} & 0 \end{pmatrix}, \\ B &= \begin{pmatrix} 0 & U_{12} \\ 0 & 0 \end{pmatrix}, \qquad [P_\Lambda, U_{\text{nl}}] = B - A = \begin{pmatrix} 0 & U_{12} \\ -U_{21} & 0 \end{pmatrix}. \end{aligned} \tag{22}$$

Table 9: Four-Pillar Structural Isomorphism Between the Ginzburg–Landau–Higgs Potential and the Bekenstein–Nyquist Shock-Core Clock Condensate.

| **Structural Pillar** | **Ginzburg–Landau / Higgs Formulation [30, 31, 32]** | **Bekenstein–Nyquist Worldline Formulation** | **Mathematical Isomorphism** |
|---|---|---|---|
| **1. Origin of Auxiliary Fiber** (Phase 5) | Perturbative unitarity restoration: scalar doublet $\Phi$ absorbs $(E/\Lambda)^2$ longitudinal growth at the Lee–Quigg–Thacker cutoff [32] | Stinespring dilation [17] (`UnitaryBandwidthDepth.lean`): $P_\Lambda U_{\mathrm{nl}} P_\Lambda$ requires $D \geq \mathrm{rank}([P_\Lambda, U_{\mathrm{nl}}])/2$ clock wires in $V_\Lambda^\perp$ | **Sz.-Nagy / Stinespring Duality:** Auxiliary scalar fiber represents the UV clock complement $V_\Lambda^\perp = (I - P_\Lambda)H$ |
| **2. Condensation Trigger** (Phase 2) | Quartic free energy: quadratic coefficient $-\mu^2$ in $V(\Phi) = -\mu^2\lvert\Phi\rvert^2 + \lambda\lvert\Phi\rvert^4$ flips from $-\mu^2 > 0$ to $-\mu^2 < 0$ | Bandwidth saturation: $\Sigma_0(x,t) = 1 - \frac{F_0}{\Lambda_{\mathrm{Bek}}\nu_{\mathrm{ML}}}$ flips from $\Sigma_0 > 0$ (smooth bulk) to $\Sigma_0 < 0$ (shock core) | **Horizon Criticality:** $-\mu^2(x,t) \equiv \Sigma_0(x,t)$; condensing $\lvert c_{\mathrm{eq}}\rvert^2 = -\Sigma_0/(2\lambda)$ restores horizon equilibrium $\Sigma(\lvert c_{\mathrm{eq}}\rvert^2) = 0$ |
| **3. Momentum-Space Locus** (Phases 2 & 4) | Defect order parameter / internal phase oscillation $\exp(-i\omega_0\tau)$ along the defect worldline | Ultraviolet Brillouin overflow ($\lvert k\rvert > \Lambda_{\mathrm{Bek}}$): $x \mapsto x + ic_{\mathrm{eff}}\tau$ rotates $\Delta k = k - \Lambda_{\mathrm{Bek}} > 0$ into internal clock winding $\exp(-i\omega_{\mathrm{sub}}\tau)$ | **UV/IR Fiber Duality:** Shifting momentum origin by $\Lambda_{\mathrm{Bek}}$ and rotating $90^\circ$ maps boundary UV overflow into fiber phase oscillation |
| **4. Worldline Dynamics** (Phases 3 & 4) | Worldline phase accumulation along defect trajectory $\gamma$ | Sub-Stepping Depth Theorem: shock worldline $\gamma$ executes $\mathcal{D}(\gamma) \geq \frac{1}{\pi\nu_{\mathrm{ML}}}\int_\gamma \lVert[P_\Lambda,[M,L]]\rVert_{\mathrm{HS}}\, ds$ sequential Givens rotations | **Action–Complexity Duality:** Proper-time phase action along $\gamma$ parallels sequential Margolus–Levitin gate depth along the causal DAG |

The formalization establishes five machine-checked theorem families (distinguishing multi-wire spatial rank gate bounds $D \geq r$ from the single-bond angle sub-stepping bound proved in `NyquistRotorGates.lean`):

1. **Single-Wire Clock Support Implies Rank-One Leakage (`rank_forwardLeakage_le_one_of_isElementaryClockGate`):** An elementary clock gate $W \in \mathrm{U}(d+m)$ satisfies `IsElementaryClockGate d m c W` if its forward leakage $(I - P_\Lambda)WP_\Lambda$ vanishes on all ultraviolet rows $\mathrm{inr}(i) \neq \mathrm{inr}(c)$ outside a single 1D clock wire $c \in \mathrm{Fin}(m)$. Factoring the non-zero row through `Fin 1` proves $\mathrm{rank}((I - P_\Lambda)WP_\Lambda) \leq 1$.

2. **One-Sided Leakage Subadditivity (`one_sided_leakage_decomposition`, `forwardLeakage_mul_rank_le` & `circuitProduct_forwardLeakage_rank_le_length`):** For any idempotent projector $P^2 = P$ and any list of unitary gates $L : \mathrm{List}(\mathrm{Matrix}\ n\ n\ \mathbb{C})$, the algebraic identity

$$(I-P)(WU)P = \big((I-P)WP\big)(PUP) + \big((I-P)W(I-P)\big)\big((I-P)UP\big) \tag{23}$$

   combined with rank subadditivity (`matrix_rank_add_le`) and induction over $L$ proves that if every $W \in L$ has $\mathrm{rank}((I-P)WP) \leq 1$, then $\mathrm{rank}((I-P)(\prod L)P) \leq \mathrm{length}(L) = D$.

3. **Mori–Zwanzig 2-Step Markovian Prediction Defect Identity and Rank Bound (`mori_zwanzig_markov_defect_eq_reverse_mul_forward` & `mori_zwanzig_markov_defect_rank_le`):** For any idempotent compression projector $P^2 = P$ and evolution operators $W, U$, the 2-step non-Markovian prediction defect satisfies

$P(WU)P - (PWP)(PUP) = (PW(I-P))((I-P)UP) = \text{reverseLeakage}(P,W) \cdot \text{forwardLeakage}(P,U)$, whose rank is bounded above by both $\text{rank}((I-P)UP)$ and $\text{rank}(PW(I-P))$, unifying Crutchfield's causal bottleneck [37, 38] and Mori–Zwanzig memory kernels [35, 36] with Stinespring clock-fiber leakage.

4. **Commutator Rank Duality** (`forwardLeakage_rank_eq_reverseLeakage_rank` & `commutator_rank_eq_two_mul_forwardLeakage_rank`)**:** Using the unitary block Gram identities $U_{11}^\dagger U_{11} + U_{21}^\dagger U_{21} = I_d$ and $U_{11}U_{11}^\dagger + U_{12}U_{12}^\dagger = I_d$ together with `Matrix.rank_conjTranspose_mul_self` ($\text{rank}(M^\dagger M) = \text{rank}(M)$), the formalization proves that forward and reverse Nyquist leakage have equal rank ($\text{rank}(A) = \text{rank}(B) = r$) and reside in orthogonal subspaces, yielding $\text{rank}([P_\Lambda, U]) = 2r$.

5. **Isometric and Block-Compression Bandwidth–Depth Tradeoffs** (`unitary_bandwidth_depth_tradeoff_compression` & `unitary_bandwidth_depth_tradeoff_clock_wires`)**:** For any target nonlinear unitary $U_{\text{nl}} \in \text{U}(d+m)$ simulated by a sequential circuit $U_D = \prod_{k=1}^{D} W_k$ of elementary 1D clock gates, both exact isometric action agreement ($U_D P_\Lambda = U_{\text{nl}} P_\Lambda$) and the weaker block-compression dilation condition ($P_\Lambda U_D P_\Lambda = P_\Lambda U_{\text{nl}} P_\Lambda$) force $\text{rank}((I-P_\Lambda)U_D P_\Lambda) = \text{rank}((I-P_\Lambda)U_{\text{nl}} P_\Lambda) = r$, establishing the unconditional multi-wire gate-count lower bound:

$$D = \text{length}(L) \geq r = \frac{1}{2}\text{rank}\big([P_\Lambda, U_{\text{nl}}]\big). \tag{24}$$

### 5.2 Extended Physics Formalizations and Upstream `Physlib` / `QuantumInfo` Bridges

To connect the abstract operator-algebraic core of `UnitaryBandwidthDepth.lean` to the physical mechanisms of Phases 2–4 and to the community Lean 4 physics ecosystem (`leanprover-community/physlib` [34], which bundles `Physlib` and `QuantumInfo`), five additional machine-checked Lean 4 modules have been constructed and verified with zero `sorry` axioms and zero compiler warnings:

1. **Concrete** SU(2) **Givens Rotors, Single-Bond Sub-Stepping, Two-Level Rabi Saturation, and Tripartite Collision Router** (`NyquistRotorGates.lean`)**:** Instantiates `IsElementaryClockGate` both on $2 \times 2$ rotors (`givensRotor_mem_unitaryGroup`, `givensRotor_isElementaryClockGate`) and non-vacuously for arbitrary spatial and clock dimensions $d, m \geq 1$ via `embeddedGivensRotor` (`embeddedGivensRotor_isElementaryClockGate`). Proves constructive in-phase clock-mass creation $G_0(\vartheta_2)G_0(\vartheta_1) = G_0(\vartheta_1 + \vartheta_2)$ (`givensRotor_in_phase_mul`), inductive $N$-step angle addition $\prod_{k=1}^{N} G_0(\vartheta_k) = G_0(\sum_{k=1}^{N} \vartheta_k)$ (`givensRotor_list_prod_in_phase`), and the single-bond ($r = 1$) sequential sub-stepping bound $\Theta \leq \text{length(angles)} \cdot \vartheta_{\max}$ (`single_bond_angle_substepping_bound`). Machine-checks anti-phase coherent clock-mass annihilation $G_\pi(\vartheta)G_0(\vartheta) = I_2$ realizing the non-monotone logical negation ($\neg$) gate primitive (`givensRotor_anti_phase_mul_eq_one`), the general Ramsey collision fringe entry $(1+\zeta)\sin\vartheta\cos\vartheta$ (`givensRotor_ramsey_fringe_entry`), two-level Rabi over-rotation reflection at $\vartheta = \pi/2$ (`rabi_overrotation_reflection`), the rank-1 channel kernel bottleneck $\dim\ker((I-P_\Lambda)WP_\Lambda) \geq (d+m) - 1$ (`rank_one_channel_ker_finrank`), strict unitarity and zero UV-spike excitation of the Milestone 4 2-stage tripartite Haar router on $(\bar{u}_b, u_{b,\text{UV}}, c_{\text{slot}})$ (`stage2MeanReadback_preserves_uv_mode`, `milestone4TripartiteRouter_mem_unitaryGroup`), and the spatial–clock Gram partition of unity $(PUP)^\dagger(PUP) + A^\dagger A = P$ (`spatial_compression_add_clock_leakage_eq_proj`).

2. **Ginzburg–Landau Shock-Core Defect Condensate Duality** (`NyquistHiggsDuality.lean`)**:** Formalizes the Bekenstein–Nyquist saturation functional $\Sigma_0 = 1 -$

$F_0/\theta_{\mathrm{Nyq}}$ (`saturationFunctional`), linear commutator defect relaxation $\Sigma(|c|^2) = \Sigma_0 + 2\lambda|c|^2$ (`saturation_linear_relaxation`), and the effective Ginzburg–Landau quartic potential $V_{\mathrm{eff}}(|c|^2) = \Sigma_0|c|^2 + \lambda|c|^4$ (`nyquistEffectivePotential`). Proves completion of the square (`HiggsPotential.complete_square`), global lower boundedness (`HiggsPotential.pos_l_lower_bound`), unique uncondensed minimization $|c_*|^2 = 0$ in the smooth sub-Nyquist bulk $\Sigma_0 > 0$ (`subNyquist_unique_minimum_zero`), spontaneous localized defect-condensate formation $|c_{\mathrm{eq}}|^2 = -\Sigma_0/(2\lambda) > 0$ in the super-Nyquist shock core $\Sigma_0 < 0$ (`superNyquist_condensate_minimum`), exact restoration of the sonic horizon $\Sigma(|c_{\mathrm{eq}}|^2) = 0$ (`superNyquist_horizon_restoration`), and cutoff-scale invariance of $\lambda$ (`quarticCoupling_cutoff_independent`).

3. **Cayley Integrator Unitarity and Gauge-Covariant Clock Parallel Transport (`NyquistGaugeTransport.lean`):** Proves that for any skew-Hermitian operator $D^\dagger = -D$ and left resolvent $R(I + D) = I$, the Crank–Nicolson / Cayley transform $U = R(I - D)$ is unconditionally unitary (`cayleyTransform_mem_unitaryGroup`). Establishes skew-Hermiticity of both the split-form Burgers advection matrix (`splitFormAdvection_skewHermitian`) and the gauge-covariant clock parallel-transport generator $D^{\mathrm{cov}}_{i,j} = w_{i,j}\sigma_{i,j}U_{i,j}$ (`covariantClockAdvection_skewHermitian`), unitary gauge equivariance $D^{\mathrm{cov}}(U') = G(\psi)D^{\mathrm{cov}}(U)G(\psi)^\dagger$ (`covariantClockAdvection_gauge_equivariant`), and exact local U(1) gauge invariance of two-shock Ramsey collision holonomy $U^{(1)}_{i,k}\overline{U^{(2)}_{i,k}}$ (`twoPathHolonomy_gauge_invariant`) and closed 3-site Wilson loops (`wilsonLoop3_gauge_invariant`).

4. **Direct Upstream `Physlib` Standard Model Higgs Potential Isomorphism (`PhyslibHiggsBridge.lean`):** Imports `Physlib.Particles.StandardModel.HiggsBoson.Potential` directly from `leanprover-community/physlib` [34] and bridges it with `NyquistHiggsDuality.lean` over smooth sections $\varphi$ : `HiggsField` on 4D `SpaceTime`. Proves pointwise identity between upstream `StandardModel.HiggsField.Potential.toFun` and `nyquistEffectivePotential` under $\mu^2 = -\Sigma_0$ (`toPhyslibPotential_toFun_eq_nyquistEffectivePotential`) and directly invokes upstream `isMinOn_iff_field_of_`$\mu$`Sq_nonpos_l_pos` and `isMinOn_iff_field_of_`$\mu$`Sq_nonneg_l_pos` to certify the sub-Nyquist uncondensed bulk ($\varphi(x) = 0$), super-Nyquist shock-core condensate ($\|\varphi(x)\|_H^2 = -\Sigma_0/(2\lambda)$), and sonic-horizon restoration (`superNyquist_physlib_horizon_restoration`).

5. **Direct Upstream `QuantumInfo` CPTP Channel & Von Neumann Entropy Bridge (`QuantumInfoClockBridge.lean`):** Imports `QuantumInfo.Channels.CPTP` and `QuantumInfo.Entropy.VonNeumann` directly from `leanprover-community/physlib` [34]. Clarifies how the single-excitation direct-sum block basis `BlockIdx` $d_{\mathrm{spatial}}$ $d_{\mathrm{clock}}$ (Section 1) embeds into the full bipartite quantum tensor product $d_{\mathrm{spatial}} \times d_{\mathrm{clock}}$ (Sections 2–3), and notes that `CPTPMap.ofUnitary` applies to fixed-angle (linear) unitary gates whereas the macroscopic Burgers lattice employs a state-dependent mean-field angle rule. Certifies our SU(2) Givens rotor and Milestone 4 tripartite Haar router as upstream unitary `CPTPMap` channels (`givensRotorCPTP_isUnitary`, `milestone4TripartiteCPTP_isUnitary`), constructs the reduced spatial fluid channel $\Phi_\Lambda(\rho_\Lambda) = \mathrm{Tr}_C(U(\rho_\Lambda \otimes \rho_C)U^\dagger)$ via `CPTPMap.traceRight` (`reducedSpatialFluidChannel_tracePreserving`), proves Stinespring purification duality via `CPTPMap.purify_trace` (`spatialChannel_admits_unitary_purification`), and formalizes Section 3.3 by proving that for any pure joint fluid+clock state $\Psi$ : `Ket`$(d_{\mathrm{spatial}} \times d_{\mathrm{clock}})$, global von Neumann entropy vanishes identically $S_{\mathrm{vN}}(|\Psi\rangle\langle\Psi|) = 0$ (`bipartite_pure_global_entropy_zero`), while the apparent macroscopic shock entropy of the reduced spatial fluid equals the entanglement entropy stored in the worldline clock fiber $S_{\mathrm{vN}}(\rho_\Lambda) = S_{\mathrm{vN}}(\rho_C) \geq 0$ (`spatial_shock_entropy_`

eq_clock_entropy) with mutual information $I(\Lambda : C) = 2S_{\mathrm{vN}}(\rho_\Lambda)$ (bipartite_pure_mutual_info_eq_two_mul_spatial_entropy).

Table 10: Summary of the Six Machine-Checked Lean 4 Formalization Modules (Zero sorry Axioms).

| Lean 4 Module | Phase / Scope | Key Formalized Structures & Theorems | Monograph Section & Physical Role |
|---|---|---|---|
| UnitaryBandwidthDepth.lean | Phase 5 (Core Mathlib) | bandlimitProj, mori_zwanzig_markov_defect, $\mathrm{rank}([P_\Lambda, U_{\mathrm{nl}}]) = 2r$, $D \geq r$ | Sections 1, 3.3 & 5.1: Stinespring dilation, Mori–Zwanzig defect, & clock depth bound |
| NyquistRotorGates.lean | Phase A (Test A1) | embeddedGivensRotor, single_bond_angle_substepping_bound, $G_\pi(\vartheta)G_0(\vartheta) = I_2$, milestone4TripartiteRouter | Sections 3.2 & 4.5: Single-bond angle sub-stepping, $\pi$-rotor $\neg$ gate, Rabi saturation, & Haar readback |
| NyquistHiggsDuality.lean | Phase A (Test A2) | saturationFunctional, nyquistEffectivePotential, $\lvert c_{\mathrm{eq}}\rvert^2 = -\Sigma_0/(2\lambda)$, $\Sigma(\lvert c_{\mathrm{eq}}\rvert^2) = 0$ | Sections 2.1 & 4.5: Localized Ginzburg–Landau shock-core defect condensate & sonic-horizon restoration |
| NyquistGaugeTransport.lean | Phase A (Test A3) | cayleyTransform_mem_unitaryGroup, covariantClockAdvection, twoPathHolonomy_gauge_invariant | Sections 2.2 & 4.1: Unconditional Cayley unitarity & U(1) Wilson holonomy invariance |
| PhyslibHiggsBridge.lean | Phase B (Upstream Physlib) | toPhyslibPotential, subNyquist_physlib_isMinOn_iff_zero, superNyquist_physlib_isMinOn_iff_condensate | Section 4.5: Direct isomorphism with StandardModel.HiggsField.Potential on 4D SpaceTime |
| QuantumInfoClockBridge.lean | Phase B (Upstream QuantumInfo) | givensRotorCPTP, reducedSpatialFluidChannel, spatial_shock_entropy_eq_clock_entropy | Sections 1.3 & 3.3: CPTPMap purification & $S_{\mathrm{vN}}(\rho_\Lambda) = S_{\mathrm{vN}}(\rho_C)$ |

# 6 Integrated Synthesis

Together, the Minimal Four-Axiom Formulation (Axioms A1–A4) and its equivalence with the Primitive Bipartite Bundle (Axioms B1–B3), the analytical proofs of Phases 1–3, the six machine-checked Lean 4 modules of Phase 5 and Phases A–B (UnitaryBandwidthDepth.lean, NyquistRotorGates.lean, NyquistHiggsDuality.lean, NyquistGaugeTransport.lean, PhyslibHiggsBridge.lean, and QuantumInfoClockBridge.lean), and the four progressive computational toy models of Phase 4 (toy_model_1_unitary_leakage.py through toy_model_4_shock_collision_gate.py, passing 45/45 automated unit, falsification, and anti-cheat assertions) establish a unified mathematical and computational foundation for the **Bekenstein–Nyquist Shock Worldline Conjecture**:

- **Unitary Dilation & Sub-Stepping Lower Bounds (Milestone 1 / Phase 5):** Any unitary simulation of a nonlinear operator across a bandlimit projector $P_\Lambda$ using 1D clock wires requires an orthogonal clock fiber $V_\Lambda^\perp$ of multi-wire gate depth $D \geq \mathrm{rank}([P_\Lambda, U_{\mathrm{nl}}])/2$ (UnitaryBandwidthDepth.lean) and single-bond sequential angle sub-stepping depth $D \geq \lceil \Theta/\vartheta_{\max} \rceil$ (NyquistRotorGates.lean).

- **Zero-Loss Unitary Shock Stabilization (Milestone 2 / Experiment H.3):** Rotating overflowing Nyquist bond modes into local worldline clock registers via impedance-matched U(2) Givens rotations ($\tan\vartheta = \sqrt{\Delta k/\Lambda_{\mathrm{Bek}}}$) eliminates Gibbs ringing while conserving total Hilbert-space norm $\|u\|_2^2 + \sum |c|^2$ to $\sim 10^{-15}$ and propagating shocks at the Rankine–Hugoniot speed $s = [f(u)]/[u]$.

- **Emergent Complexity Phase Transition & Asymptotic Tightness (Milestone 3 / Experiment H.4):** Purely state-driven $\mathcal{H}_{2\Lambda}$ spectral spillover detection yields $\mathcal{O}(1)$ parallel circuit depth for $t < t^*$ and transitions at $t \approx t^*$ to linear sequential depth $\Omega((T - t^*)\Lambda_{\mathrm{Bek}})$ along 1D shock worldlines, converging to within 0.51% of the unrounded discrete gate-sum bound at $\Lambda_{\mathrm{Bek}} = 256$.

- **Holographic Bulk Extrusion & Ryu–Takayanagi Min-Cut Correspondence (Subsection 4.4.1):** Ultraviolet Bekenstein–Nyquist overflow on the spatial boundary $\mathcal{H}_\Lambda$ dynamically extrudes sequential depth in the orthogonal clock fiber $\mathcal{H}_{\mathrm{clock}}$ and warps the effective horizon metric $g_{00} \leftrightarrow \Sigma(x,t)$, providing an operator-algebraic realization of Ryu–Takayanagi min-cut wire counting along $\gamma$ [28, 29].

- **Coherent Clock-Mass Readback & Ginzburg–Landau–Higgs Isomorphism (Milestone 4 / Phase 3.2):** Coupling a co-moving worldline clock register to the orthogonal 2-site Haar basis $(\bar{u}_b, u_{b,\mathrm{UV}})$ realizes both the unitary $\pi$-rotor logical negation ($\neg$) gate primitive required to evade the Monotone Planar CVP $\mathrm{NC}^3$ trap [24] and a 77.02% coherent clock-mass annihilation fringe across $\Delta\phi \in [0, 2\pi]$.

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